

BOB-096 “ Chaos coverage for qBittorrent-go/tests/ integration/jackett_db_test.go

Ticket: BOB-096 **Scope:** extend the boba-jackett DB integration suite with §11.4.85-class process-death, resource-exhaustion, and concurrent-kill chaos scenarios. **Date:** 2026-08-19 **Race detector:** clean (go test -race, 3— deterministic re-run).

Scenarios added

#	Scenario	Test function	§11.4.85 class
	Resource-exhaustion “ file-descriptor		
1	pressure via os.Pipe() (bounded per §12.6)	TestChaos_FileDescriptorExhaustion	(d)
	Concurrent-kill via context.CancelFunc		
2	from a second goroutine while the first is writing	TestChaos_ConcurrentContextCancelWrite (a-extra)	

Scenarios already present (kept, referenced for completeness)

#	Scenario	Test function	§11.4.85 class
	Process-kill (SIGKILL of a helper subprocess mid-transaction)		
3		TestChaos_MidTransactionSIGKILLRecovery	(a)
	Byte-level		
4	DB file corruption	TestChaos_DBFileByteCorruption	(b)
	Concurrent-writer contention (40 goroutines,		
5		TestChaos_ConcurrentWriterContentionRepo	(c)

#	Scenario	Test function	Â§11.4.85 class
	direct repo layer)		
	WAL sidecar		
6	corruption + reopen	TestChaos_WALSidecarCorruptionRecovery	(a-analogue)
	Master-key rotation		
7	mid-flight (K1â†'K2 forward + K1 reverse)	TestChaos_MasterKeyRotationMidflight	(e-analogue)

Â§11.4.115 RED-first evidence â€” captured live

For `TestChaos_ConcurrentContextCancelWrite`, invariant (ii) (â€œevery confirmed write round-tripsâ€”) was proven falsifiable by injecting a phantom name `KILL_PHANTOM` into `finalNames` with no corresponding DB row.

- `RED_phantom_mutation.log` â€” captures the FAIL: (ii) confirmed-write `KILL_PHANTOM` lost after cancel: repos: credential not found â†' FAIL: `TestChaos_ConcurrentContextCancelWrite`
- Mutation reverted; `GREEN_after_restore.log` captures the immediate PASS under identical race conditions.

Falsification narratives for the other invariants (i, iii on FD + cancel tests; all invariants on the pre-existing tests) are recorded in the per-test doc-comments per Â§1.1.

Anti-bluff assertions (Â§11.4.5 / Â§11.4.107)

- Byte-equal plaintext round-trip via `Credential.Username/Password`, not â€œno errorâ€”.
- `sql.DB` row count vs writer-confirmed count exact equality.
- Categorised error on FD-pressure Upsert failure (non-empty `err.Error()`).
- Durability boundary: BASELINE / PRECOMMIT rows always survive; uncommitted rows never surface.
- Panic guard: every DB access under chaos wrapped in `recover()` â†' `t.Fatalf`.

Cleanup (Â§11.4.14)

- `TestChaos_FileDescriptorExhaustion` closes every `os.Pipe()` pair via both `defer cleanup()` and `t.Cleanup(cleanup)` (belt-and-braces).

- `TestChaos_ConcurrentContextCancelWrite` registers `t.Cleanup(cancel)` so the context is cancelled even on early test bailout.
- `t.TempDir()` handles DB + evidence-log cleanup automatically.

Host safety (Â§12.6, Â§12.12, Â§11.4.161)

- FD pressure capped at `min(RLIMIT_NOFILE.soft / 4, 512)` â€” never destabilises the host FD table.
- No sudo, no rootful container, no host power-management calls.
- `GOMAXPROCS=2 nice -n 19 ionice -c 3` throughout.

Determinism (Â§11.4.50)

Three consecutive `-race -count=3` runs â€” all GREEN (`race_detector_3x.log`).

Run log files

- `bob096_run_verbose.log` â€” verbose PASS output for the two new tests.
- `full_chaos_suite.log` â€” full `TestChaos*` suite under `-race`.
- `new_chaos_scenarios.log` â€” first run of the two additions.
- `RED_phantom_mutation.log` â€” captured RED (invariant (ii) fails).
- `GREEN_after_restore.log` â€” GREEN immediately after restoring source.
- `race_detector_3x.log` â€” 3Ã— deterministic race-clean run.

Skipped chaos class (honest, tracked)

Â§11.4.85 (d) `disk-full` remains deferred as a follow-up Â§11.4.197 item â€” a `filesystem-quota` fixture belongs in a container-scoped harness rather than a host-shared test file. Recorded in the test-file doc-comment.